

Module 4 Homework I

 Publish

Java Swing Database Program

Objective:

To create a Java Swing application for managing student records and courses in a MariaDB database. The application should allow users to add, edit, and update existing records in two tables.

Prerequisites:

- Basic understanding of Java programming.
- MariaDB installed on the local machine.
- MariaDB JDBC driver included in the project.

Instructions:

Part 1: Database Setup

1. Install MariaDB:

- Install MariaDB on your machine.
- Create a database named `University`.

2. Create Tables:

- Execute the following SQL queries to create the necessary tables:

```
CREATE TABLE Students ( studentID INT PRIMARY KEY, firstName VARCHAR(50), lastName VARCHAR(50) ); CREATE TABLE Courses ( courseID INT PRIMARY KEY, studentID INT, courseTitle VARCHAR(100), CONSTRAINT fk_student FOREIGN KEY (studentID) REFERENCES Students(studentID) );
```

Part 2: Java Swing Application

3. Download JDBC Driver:

- Download the MariaDB JDBC driver from <https://mariadb.com/>  [\(https://mariadb.com/\)](https://mariadb.com/).
- Add the JAR file to your project.

4. Create Java Swing Application:

- Use the provided Java code for the Swing application.
- Modify the database connection details in the code.

5. Add Record Button:

- Implement the logic for the "Add Record" button.
- When clicked, the application should prompt the user to enter a new student's ID, first name, and last name. It should then insert this record into the `Students` table.

6. Delete Record Button:

- Implement the logic for the "Delete Record" button.
- When clicked, the application should prompt the user to enter the student ID to delete and it should then delete the corresponding record in the `Students` table.

7. Update Existing Record Button:

- Implement the logic for the "Update Existing Record" button.
- When clicked, the application should prompt the user to enter the student ID whose record needs to be updated. It should then update the corresponding record in the `Students` table.

8. Exception Control:

- Implement exception control to handle potential errors gracefully.
- Provide informative error messages.

9. Display Record

- Implement display record to show existing records.

Grading Rubric:

- **Add Record (30 points):**

- Correctly inserts a new record into the `Students` table.
- Handles input validation.

- **Edit Record (30 points):**

- Correctly updates an existing record in the `Students` table.
- Handles input validation.

- **Update Existing Record (30 points):**

- Correctly updates an existing record in the `Students` table using a separate button.
- Handles input validation.

- **Exception Control (5 points):**

- Effectively implements exception control to handle potential errors.
- Provides informative error messages.

- **UI and Usability (5 points):**

- The application should have a clear and user-friendly interface.
- Properly formatted text fields and buttons.

Submission:

- Submit the modified Java code.
- Include at least two screenshots demonstrating the application's functionality (adding, editing, and updating records).

Points 100

Submitting a file upload

Due	For	Available from	Until
Nov 24, 2025	Everyone	-	Dec 5, 2025 at 11:59pm